Choice 1: the module skin, cut BT80 or printed

The **skin** is the outer surface of the swappable module. In V4 it is a real BT80 tube (outside diameter 66.04 mm) with four windows cut in it; each cut-out piece is glued to a fin so the fin closes flush with the tube. For the modular design the module has to be rebuilt after every rough flight, so how the skin is made decides how long a swap takes. Everything below is a concept drawn from the V4 model; the printed-skin numbers are my calculations, not a CAD result.

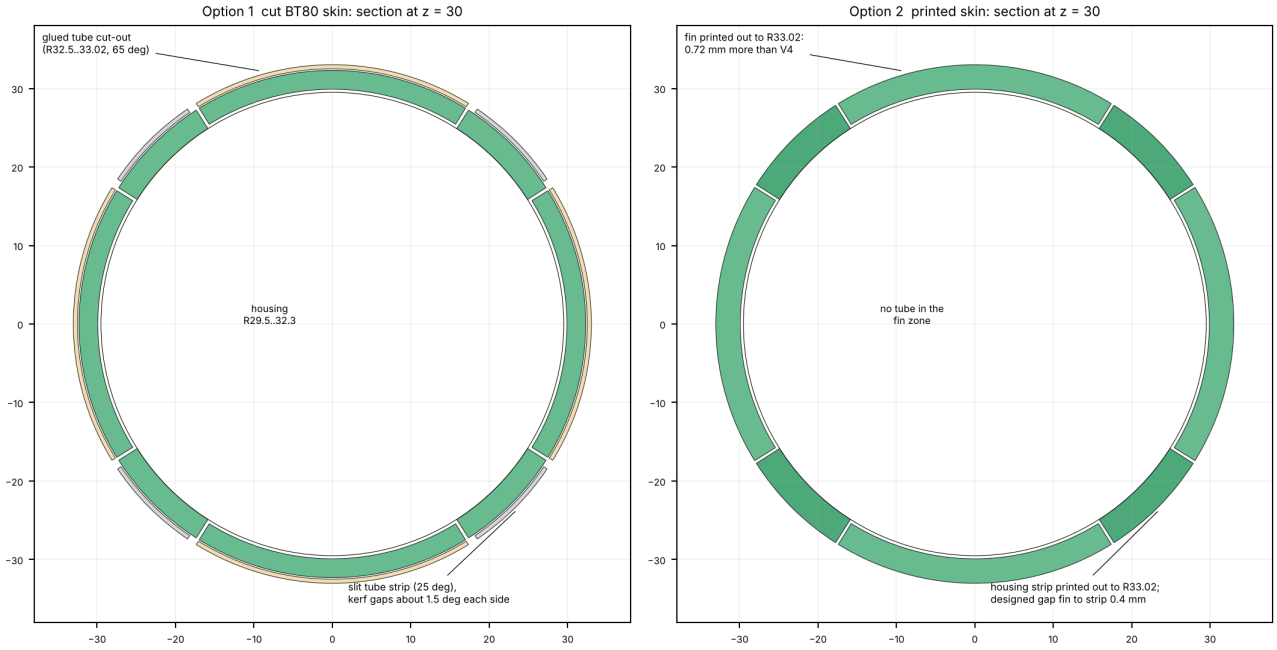
1 The two options

	Option 1: cut BT80 skin (as V4)	Option 2: printed skin
What it is	A 92.8 mm piece of real BT80 with four windows 37.5 × 42 mm. The four cut-outs are glued to the fins. The skin is screwed to the housing.	The housing strips and the fins are printed out to the BT80 outside diameter, R33.02 (0.72 mm more than V4). No tube in the fin zone; the real BT80 above and below butts against it.
What you do per module	Cut a tube piece to length, mark and cut four windows (template supplied), cut-outs glued to the fins, wait for the glue, screw the skin on.	Print the module. Optionally sand and paint. No cutting, no glue, no cut-outs to line up.
Why it can be better	The surface is the real tube, smooth and the right diameter; no extra PLA; the tube is cheap and strong.	Repeatable and fast to rebuild; no glue joint between fin and cut-out (that joint is a place a fin can fail); the walls get thicker (strips 3.5 mm, fins 3.1 mm) so the module is stiffer.
Why it can be worse	Hand cutting four 65° windows to a sub-millimetre fit takes time and the gap varies; glue needs cure time; every module consumes a tube piece and a gluing session.	About +17 g of PLA per module (see the numbers); layer lines need sanding for a smooth finish (drag not measured); the printed OD may be ±0.2 mm, which matters where it butts the real tube.
Works with which fastening	Screws only: a 0.52 mm tube wall cannot take a recess or a bayonet slot.	Screws, or thumb screws, or a bayonet (see the fastening document).

2 The numbers

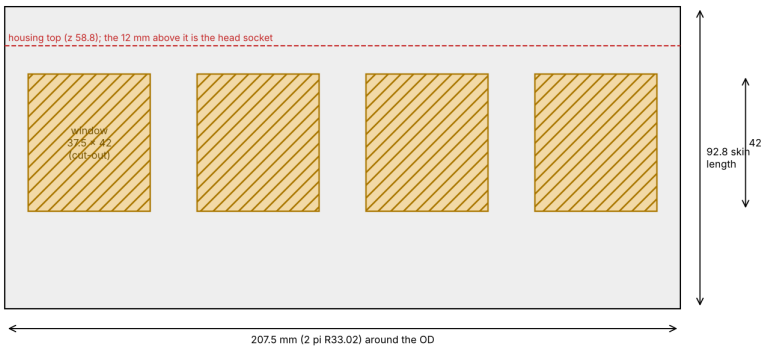
Item	Option 1	Option 2
Skin outside diameter	66.04 mm (real tube)	66.04 mm (printed, same)
Skin length	92.8 mm (z –22 … 70.8: 12 mm base sleeve, mechanism, 12 mm head socket)	92.8 mm
Windows / fin interface	four 37.5 × 42 mm cut-outs, glued; gaps equal to the saw kerf (assume 1.5° each side)	designed gap 0.4 mm all round; the fin carries its own skin
Wall at the fin zone	tube 0.52 + fin shell 2.4 (two parts joined by glue)	one printed part 3.1 mm (fin), 3.5 mm (strip)
Extra printed mass	0 g (the tube is a separate purchase: weigh yours)	about +17 g per module: $\pi \times (33.02^2 - 32.3^2) \times 92.8$ mm × 1.24 g/cm³ (module about 122 g instead of 105 g)
Hands-on work per module	cutting plus gluing plus cure time (not timed: my guess is tens of minutes plus the cure)	none beyond optional sanding; print time is unattended
Risk	fit of the cut-outs; glue failure	printed OD tolerance; surface finish; the first print will need a fit check

3 Drawings

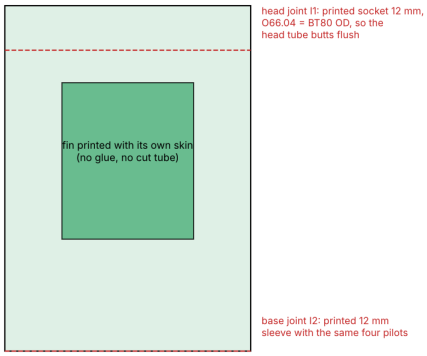


Section at z = 30 through the fin zone. Left: V4 style, real tube strips and glued cut-outs outside the fins and strips. Right: printed skin, fins and strips printed out to the BT80 outside diameter.

Option 1: the slit skin, developed flat, starting at the middle of a strip (1:1 template in its own file). Windows 37.5 × 42, centres 51.9 mm apart, strips 14.4 mm



Option 2: the printed module skin, elevation. Outside diameter 66.04 mm all the way, so it mates with the real BT80 above and below



Top: the Option 1 skin developed flat (window sizes and spacing). Bottom: the Option 2 module outline with the head and base joints, outside diameter 66.04 mm all the way.

4 What I recommend

- **Go with the printed skin (Option 2) for the module**, because the module is the part you will rebuild after every rough flight and cutting and gluing four fin cut-outs each time is the slowest, most variable step. It also lets the swap joints be built into the printed module (next document).
- **First test:** print one fin and a 60 mm skin sector and check three things: the fin-to-strip gap closes without binding, the printed outside diameter (66.04 ±0.2) against a real BT80 end, and the surface after sanding. Do this before printing a whole module.
- **Keep Option 1 available.** The 1:1 cutting template (a separate file) lets you build a V4-style skin from a real tube in the meantime; it is derived from the V4 windows and should be checked against your own tube before cutting.
- **Not verified:** drag of a printed surface against the tube, the effect of the extra 17 g on the rocket's balance and apogee, and whether the printed fin skin cracks at the hinge ears on a hard landing (it is thicker than V4, so it should not be worse, but nothing has been tested).